JINHO CHOI

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SUMMARY

Ph.D. candidate at KAIST and visiting researcher at Helmholtz Munich, working on **mechanistic interpretability** and **data-centric AI** to advance **safe and trustworthy AI**. Ph.D. research uses **sparse dictionary learning** to uncover human-interpretable concepts in foundation models: interpreting vision models (PatchSAE), uncovering dataset biases (ConceptScope), analyzing vision-language models (VisualScratchpad), and improving the consistency of learned concepts across training runs (ConSDL). Five years in industry as an AI scientist, developing biomedical imaging models at Tomocube and AI interview systems and conversational agents at Genesislab. M.S. research in visual analytics and human-computer interaction (HCI).

EDUCATION

- **Ph.D. in Artificial Intelligence**, KAIST (Advisor: Jaegul Choo), Aug. 2022 – Feb. 2028 expected, part-time 2022–2024
- **M.S. in Computer Science and Engineering**, Korea University — Advisor: Jaegul Choo, Mar. 2017 – Aug. 2019
- **B.S. in Computer Science and Engineering**, Korea University, Mar. 2013 – Feb. 2017

EXPERIENCE

Helmholtz Munich **Feb. 2026 - Present**
Visiting Researcher, Dynamical Inference Lab *Munich, Germany*
- Developed **ConSDL**, a sparse coding approach that improves the consistency of learned concepts across independent training runs. Demonstrated improved dictionary and coefficient consistency over six sparse autoencoder baselines on vision and language models [11].

KAIST **Aug. 2022 - Present**
Ph.D. Researcher, DAVIAN Lab *Daejeon, Korea*
- Developed **ConceptScope** to characterize image datasets through interpretable visual concepts, revealing previously overlooked dataset biases [9].
- Applied sparse autoencoders to CLIP (**PatchSAE**) to study visual concept representations and their changes during task adaptation [8]. Extended concept-level analysis to large vision-language models with **VisualScratchpad** [10].

Genesislab **Sep. 2022 - Nov. 2024**
AI Scientist *Seoul, Korea*
- Developed a **fairness diagnosis and mitigation pipeline** for an AI video-interview platform used by 150+ companies, reducing gender bias by 35% without sacrificing accuracy [6].
- Developed a **talking-face generation** model deployed as an AI interviewer avatar, with accurate lip synchronization and high visual quality.
- Built **LLM-based conversational agents** for a creator-persona app with 500K+ downloads, using retrieval-augmented generation, memory, and automated persona construction.

Tomocube **Feb. 2019 - Sep. 2022**
AI Researcher *Seoul, Korea*
- Developed **3D cell and organelle segmentation** models for label-free holotomography, integrated into commercial cell analysis software [4, 5].
- Built a **human-in-the-loop 3D annotation system** that propagates 2D labels into 3D and requests corrections on uncertain slices, and deployed it as the company's internal annotation system [7].

Korea University **Mar. 2017 - Aug. 2019**
M.S. Researcher *Seoul, Korea*
- Developed a **chart accessibility tool** that extracts data from web-based visualizations and makes it accessible to visually impaired users through screen readers [3].
- Developed **visual analytics systems** for detecting anomalous local events in geo-tagged social media using topic modeling and interactive map visualizations [1, 2].

PUBLICATIONS

[11] **Jinho Choi**, Hyesu Lim, Jaegul Choo, and Steffen Schneider. Beyond SAEs: Consistent Concept Discovery with Sparse Coding. *Under review*, 2026.

[10] Hyesu Lim, **Jinho Choi**, Taekyung Kim, Byeongho Heo, Jaegul Choo, and Dongyoon Han. VisualScratchpad: Grounding Visual Concepts in Large Vision Language Models. *ICLR 2026 Workshop on Trustworthy AI*, 2026. [pdf] [code] [project]

[9] **Jinho Choi**, Hyesu Lim, Steffen Schneider, and Jaegul Choo. ConceptScope: Characterizing Dataset Bias via Disentangled Visual Concepts. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [pdf] [code] [project]

[8] Hyesu Lim, **Jinho Choi**, Jaegul Choo, and Steffen Schneider. Sparse Autoencoders Reveal Selective Remapping of Visual Concepts During Adaptation. *International Conference on Learning Representations (ICLR)*, 2025. [pdf] [code] [project]

[7] Wonwoo Cho, Dongmin Choi, Hyesu Lim, **Jinho Choi**, Saemee Choi, Hyun-seok Min, Sungbin Lim, and Jaegul Choo. Slice and

Conquer: A Planar-to-3D Framework for Efficient Interactive Segmentation of Volumetric Images. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024. [pdf]

[6] Changwoo Kim, **Jinho Choi**, Jongyeon Yoon, Daehun Yoo, and Woojin Lee. Fairness-aware Multimodal Learning in Automatic Video Interview Assessment. *IEEE Access*, 11:122677–122693, 2023. [pdf]

[5] **Jinho Choi**, Hye-Jin Kim, et al. Label-free Three-dimensional Analyses of Live Cells with Deep-learning-based Segmentation Exploiting Refractive Index Distributions. *bioRxiv*, 2021. [pdf]

[4] **Jinho Choi**, Junwoo Park, Hyun-seok Min, Hyungjoo Cho, Sungbin Lim, and Jaegul Choo. 3D Cell Instance Segmentation via Point Proposals using Cellular Components. *SPIE Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues XIX*, 2021. [pdf]

[3] **Jinho Choi**, Sanghun Jung, Deokgun Park, Jaegul Choo, and Niklas Elmqvist. Visualizing for the Non-Visual: Enabling the Visually Impaired to Use Visualization. *Computer Graphics Forum (EuroVis)*, 38(3):249–260, 2019. [pdf]

[2] Minsuk Choi, Sungbok Shin, **Jinho Choi**, et al. TopicOnTiles: Tile-based Spatio-Temporal Event Analytics via Exclusive Topic Modeling on Social Media. *ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2018. [pdf]

[1] Sungbok Shin, Minsuk Choi, **Jinho Choi**, et al. STExNMF: Spatio-Temporally Exclusive Topic Discovery for Anomalous Event Detection. *IEEE International Conference on Data Mining (ICDM)*, 2017. [pdf]

SERVICE

- **Reviewer:** NeurIPS, ICLR

SKILLS

- **Technical:** Python | PyTorch | Lightning | Hugging Face | Hydra | SLURM / HPC | CUDA | Libtorch | Git | Linux
- **Languages:** Korean (native) | English (professional)